

ViviClay: Fabricating Ceramics with Animated Surface Effects

Jingxin Ye*
Intelligent Big Data Visualization Lab
Tongji University
Shanghai, Shanghai, China
2333342@tongji.edu.cn

Qiaoqiao Jin*
Tongji University
Shanghai, China
eventualli@163.com

Chao Yuan
Shanghai Research Institute for
Intelligent Autonomous Systems
Tongji University
Shanghai, China
2210992@tongji.edu.cn

Xuechen Li
Intelligent Big Data Visualization Lab
Tongji University
Shanghai, China
primav1022@tongji.edu.cn

Yang Shi
Intelligent Big Data Visualization Lab
College of Design and Innovation,
Tongji University
Shanghai, China
shiyang1230@gmail.com

Qing Chen
Intelligent Big Data Visualization Lab
Tongji University
Shanghai, China
qingchen@tongji.edu.cn

Nan Cao
Intelligent Big Data Visualization Lab
Tongji College of Design and
Innovation
Shanghai, China
nan.cao@gmail.com

Guanhong Liu†
College of Design and Innovation,
Intelligent Big Data Visualization Lab
Tongji University
Shanghai, China
liugh23@tongji.edu.cn



Figure 1: A ViviClay couple cup shows a heart animation when hot water is poured in. The hot water is drinkable and flows through the channels into the cup for drinking.

Abstract

We present ViviClay, an innovative ceramic material that integrates thermochromic coatings and internal pipe structures to enable dynamic surface effects. By channeling thermal fluid through these printed pathways, the ceramic surface undergoes reversible color changes, creating animated patterns that are more diverse and

computable than those of conventional thermochromic mugs. To address challenges in ceramic 3D printing, we propose design strategies, optimize key parameters, and introduce a relief-style printing method for curved surfaces. This work represents the first exploration of 3D-printed ceramics enabling animated effects on object surface within the HCI field, offering ceramic users interactive and expressive new product possibilities.

*Both authors contributed equally to this research.

†Corresponding author.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719685>

CCS Concepts

• **Human-centered computing** → *Interactive systems and tools*.

Keywords

Clay 3D Printing, Ceramics, Thermochromic Materials, Dynamic Display

ACM Reference Format:

Jingxin Ye, Qiaoqiao Jin, Chao Yuan, Xuechen Li, Yang Shi, Qing Chen, Nan Cao, and Guan hong Liu. 2025. ViviClay: Fabricating Ceramics with Animated Surface Effects. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25), April 26–May 01, 2025, Yokohama, Japan*. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3706599.3719685>

1 Introduction

Dynamic displays and color-changing interfaces is widely researched in HCI field. A recent trend focuses on integrating these technologies into everyday materials such as paper [12, 23], plastic [13, 15], and fabric [1, 21, 22]. Although dynamic display technologies have been widely applied to various materials, the exploration of dynamic displays on ceramic surfaces is still in its early stages. Ceramics play a crucial role in both daily life and culture, offering a unique combination of functionality and artistry. Their distinct material properties and diverse forms make them ideal for practical use, aesthetic expression, and cultural heritage. Introducing dynamic displays to ceramic surfaces has the potential to expand their design language and visual expressiveness, enhance user interaction, and increase the market value and commercial applications. HCI research on clay 3D printing has primarily focused on optimizing printing quality [5, 8, 25], exploring innovative printing materials [2, 3, 9], and enhancing interactive experiences [6, 11, 16]. However, the research of ceramics' dynamic display capabilities remains to be further explored.

In this paper, We introduce ViviClay—a novel ceramic material capable of producing animated effects on physical surfaces. We explored printing pipe structures within ceramic walls, using injected thermal fluid to drive temperature and color changes in the thermochromic coating on the ceramic surface, thereby creating animated effects. As a late-breaking work, we hope ViviClay will lay the foundation for future research on dynamic display about ceramics. We envision its application in water-related scenarios, such as drinking cups, bathroom decorations, and public water feature designs, to enhance the aesthetic expression and functional value of ceramic artworks through its dynamic animation effects.

The contributions of this paper includes:

- 1) This paper explores fabricating animated surface effects on ceramics material for the first time. Unlike commercially available thermochromic mugs limited to bottom-up color changes, this study achieves richer, more computable dynamic effects and information display.
- 2) An exploration of the challenges involved in printing pipe structures on both flat and curved ceramic surfaces, along with corresponding design strategies and printing parameter recommendations, especially a relief-style printing method that enables successful printing of pipe structures on curved surfaces.
- 3) Establish a design space for animation effects applicable to both flat and curved surfaces, and a cup-based application example.

2 Background and Related work

With advances in materials and HCI technologies, researchers are exploring dynamic displays and color-changing interfaces on everyday materials, such as paper [12, 23], plastics [13, 15], and fabrics [1, 21, 22]. These technologies rely on materials that respond to environmental changes, exhibiting color effects under

various driving mechanisms. Common mechanisms include thermochromic [1, 12, 13, 15, 21–24], photochromic [10, 19, 20], electrochromic [7, 17], and fluid-based [4, 14] mechanisms.

However, ceramics, a common material for decoration and daily use, have been scarcely explored as objects capable of displaying animated effects. Current HCI research on clay 3D printing primarily focuses on the following aspects: (1) Optimizing printing quality, which involves refining printing paths with specialized software to enable the precise printing of complex geometries [5, 8, 25]. (2) Exploring innovative printing materials, where researchers have investigated various approaches to enhance material properties and expand the design possibilities of clay 3D printing. Examples include mixing clay with biomaterial dough [2] and embedding filaments in extruded clay to enhance tensile strength, enabling the printing of complex overhangs and unsupported structures [9]. (3) Enhancing the interactive experience, where studies explore methods such as sketching paths [6], using joysticks [11], or employing visualization panels [18], allowing users to directly control the printing process.

To drive the surface of ceramics to exhibit animated effects, Thermotion has provided us with great inspiration [26]. Thermotion used 3D printing technology to fabricate piping structures within plastic, applied a heat diffusion layer on the surface, and coated it with thermochromic paint. Finally, heat fluid was introduced into the piping structure to achieve dynamic color-changing effects. Ceramics have hydrophilic properties and are often used for containers that hold water. If fluid channels could be printed inside ceramics and coated with thermochromic paint on the surface, dynamic effects could be observed when water is poured inside, enhancing the visual expressiveness of ceramic artworks.

Nevertheless, we cannot simply apply Thermotion's technology to ceramics. This is because clay is in a semi-fluid state during printing, making it prone to collapse and clogging, which makes it difficult to print fine, complex channels like those used in Thermotion's design. Therefore, the design possibilities and fabrication processes that clay materials can support are still unknown. In this late-breaking work, we will conduct preliminary research on the fabrication process and design space of ceramics with animated surface effects.

3 ViviClay Composites

3.1 Overview of ViviClay Mechanism and Fabrication Pipeline

As shown in Figure 2-a, the core mechanism of achieving animation effects on the ceramic surface is to utilize the thermochromic paints' response to temperature variations. Specifically, we designed and printed ceramic objects with pipe structures inside walls. After firing, we apply thermochromic paints to the ceramic surface. When hot fluid flows through the pipes, the temperature change is transferred to the ceramic surface. Upon reaching a certain temperature threshold, the thermochromic paints undergoes a reversible color change. The ceramic surface can displays various dynamic textures on the different pipe pathways.

The fabrication process of ViviClay consists of the following five key steps in Figure 2- b:

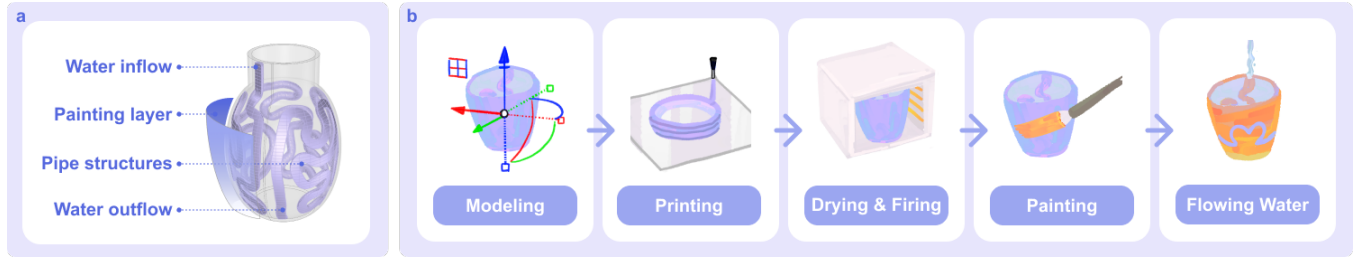


Figure 2: Overview of ViviClay. (a) The mechanism and structure of ViviClay. (b) Fabrication Pipeline

- (1) **Modeling**: this step involves designing and modeling objects with pipe structures in their side walls. For flat and curved surfaces, we designed two set of printing methods, as mentioned in Section 3.2.
- (2) **Printing**: this step involves using FDM clay 3D printing technology to fabricate the objects.
- (3) **Drying and Firing**: printed objects must be air-dried to remove moisture and then fired at high temperatures to solidify into ceramics. Before firing, glaze can be applied to paint decorative patterns on the surface.
- (4) **Painting**: this step involves applying thermochromic paint to the surface of the ceramic. Additionally, a layer of transparent protective coating is applied to ensure the long-term stability and effectiveness of the paint.
- (5) **Flowing Water**: this step involves flowing hot water through the pipes to activate dynamic textures on the surface of ViviClay.

3.2 Key Printing Parameters and Structural Experiments

Clay remains in a semi-fluid state during printing. When printing pipe structures in the side walls of an object, issues such as the collapse and deformation of the clay coil often occur, leading to pipe blockages. If these blockages occur, fluids cannot flow through the pipes, preventing the realization of color-changing dynamic effects. To address these challenges, we explored key printing parameters and pipe structures on both flat and curved surfaces.

3.2.1 Flat Surfaces. In the flat surface printing process, we utilized a relatively narrow extrusion head with a line diameter of 1.2mm and a layer thickness of 0.4mm for our experiments. We tested four different pipe shapes—square, circle, triangle, and arch, as illustrated in Figure 3. To evaluate the printing performance of different pipe shapes while ensuring consistent object appearance and animation effects, we set the maximum width and height of each pipe to 4mm. A consistent maximum width ensures uniform texture line width, while a consistent height ensures uniform overall thickness of the flat object.

From the top view, the square pipe was observed to lack a narrowing structure at the top, which caused the upper clay coil to drop significantly during the printing process. This issue rendered the square shape unsuitable for creating a closed pipe structure. In contrast, the circular, triangular, and arch pipes demonstrated considerably fewer problems, making them more viable options for closed structures.

Examining the side view, it became evident that the semi-fluid state of the clay during extrusion led to varying degrees of shrinkage and deformation across all pipe shapes after printing. This deformation is a critical factor to consider, especially since clay undergoes further shrinkage during the firing process. To ensure smooth water flow within the pipes, we compared the cross-sectional areas of the remaining three shapes—circle, triangle, and arch. Under identical maximum width and height conditions, the arch pipe maintained a larger cross-sectional area compared to the others. Therefore, we finally chose the arch pipe as the optimal shape.

3.2.2 Curved Surfaces. When printing pipes on curved surfaces, we initially attempted to print the pipes within the curved surface. During the experiments, we frequently encountered issues where the upper clay layers failed to attach properly and dropped, particularly when printing certain pipes, such as horizontal ones (Fig. 4-a). Since the pipes were relatively narrow, any deformation or collapse of the pipe structure often led to blockages. Additionally, considering the shrinkage of clay after drying and firing, we had to design the radius of the pipe to be large enough. However, adding an external shell made the curved surface excessively thick, which was unsuitable for certain applications. To address these challenges, we introduced a relief-style printing method (Fig. 4-b). Specifically, we designed an arched and protruding pipe structure on the curved surface. The wall thickness was set to twice the nozzle diameter, with zero infill, printing only the inner and outer walls. The outer wall forms a smooth surface, while the inner wall forms the arched, protruding pipe. The smooth outer surface will be coated with thermochromic paints to display animation effects, while the inner surface will not be used for animation presentation.

While this method can successfully print unobstructed pipelines, several challenges were encountered during the printing process. Firstly, if the starting point of each layer remains fixed at the same position, noticeable seams may form at the joints, potentially leading to fractures (Fig. 4-c). To address this, it is necessary to randomize the starting point of each layer during slicing. Additionally, when printing horizontal pipes, the elevated arch-shaped pipes are prone to deformation during the printing process due to their steep inclination and the semi-fluid state of clay. This deformation can lead to misalignment and fractures, ultimately preventing the successful printing of upper layers (Fig. 4-d). In contrast, flatter arched pipes demonstrated improved printability and better overall results (Fig. 4-e).



Figure 3: Different pipe shapes and their printing conditions.



Figure 4: Experiments on printing pipes on curved surfaces. Two different methods for printing pipes: (a) printing pipes within the curved surface, which proved unfeasible. (b) Printing pipes using a relief-style method. Challenges encountered during the relief-style printing process: (c) fixed layer start points may cause visible seams at joints. (d) A steep angle at the highest point of elevated arched channels makes them prone to deformation and fracture during printing. (e) Suitable pipe form and parameters for the relief-style printing method.

4 Design Space

4.1 Animation Effects Primitives

4.1.1 Animation Effects on Flat Surfaces. As shown in Figure 5, we categorize animations on flat surfaces based on the number of paths into one-stroke and multi-stroke. For the one-stroke path, it is further divided by its effects into semantic icon and expressive textures. Semantic icons are further categorized into those “with intersections” and “without intersections”. For the multi-stroke path, there is only texture effects.

4.1.2 Animation Effects on Curved Surfaces. As shown in Figure 6, we construct the design space of curve surfaces based on three factors: contour lines, generatrix, and whether the contour lines twist around the center. Among these, contour lines are categorized into circular and non-circular lines. Generatrix are classified as straight lines or curved lines. “Twist or not” refers to whether non-circular contour lines rotate around the center to form a twisted curve surface. By combining these three factors, we derived six types of curve surfaces. Each type of curve surfaces can be combined with a single-stroke path to create different dynamic patterns.

4.2 Animation Effects with Different Painting Methods

The thermochromic paints are available in two types: colorless to colored and colored to colorless. Creators can choose between these two types of paints and apply them in the following two ways:

- **Flat Painting.** The thermochromic paints are evenly applied across the entire ceramic surface (Fig. 7-a & b). This approach allows

the ceramic surface to display dynamic displays that correspond to the pipe path shapes when water flows through.

- **Patterned Drawing.** Thermochromic paints and glazes are used to create patterns on the ceramic surface (Fig. 7-c). Creators can draw arbitrary patterns with glaze on the ceramic surface, and thermochromic paints are then applied to specific areas after firing. The strategic combination of glaze patterns and thermochromic paints enables a more expressive presentation of temporal dynamic effects.

4.3 Assemble Design space into Compound Shapes

Due to the molding limitations of clay materials during the printing process, the clay remains in a semi-fluid state, making it prone to collapse and deformation. If one intends to print complex channel structures on an object’s surface using a monolithic fabrication method, the design space is confined to the range shown in Figure 6. Currently, the relief-style printing method is only suitable for printing simple pipeline structures on the surfaces of three-dimensional objects that feature smooth and gentle curves, such as cups and vases. To further expand the possibilities of ceramic shapes, we introduced an assembly process. By combining the design space of flat surfaces (Figure 5) with diverse hand-crafted or 3D-printed components, one can fabricate compound shapes with dynamic textures. Flat surfaces with internal pipes can serve as essential elements of three-dimensional objects—such as cup walls, bases, or decorative features—as demonstrated in the case study in Section 5.

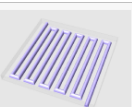
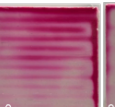
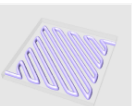
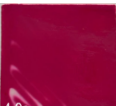
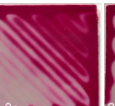
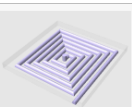
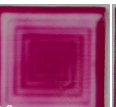
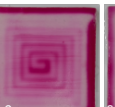
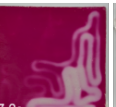
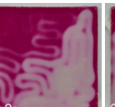
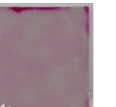
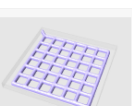
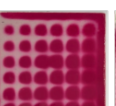
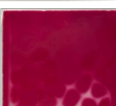
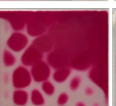
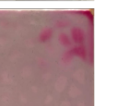
Path type	Pattern type	3D model & Specification		Animation			
Single-stroke path refers to pipe structures that allow water to flow from the inlet to the outlet along a single continuous path.	Semantic Icon without intersections		Butterfly Size: 10.0*10.0 cm Wall thickness :1.2mm Duration: 10.4s				
			Heart Size: 10.0*10.0 cm Wall thickness :1.2mm Duration:12.0s				
			Direction Size: 10.0*10.0 cm Wall thickness :1.2mm Duration: 11.0s				
	Semantic Icon with intersections		Bulb Size: 10.0*10.0 cm Wall thickness :1.2mm Duration: 10.3s				
			Horizontal wiping Size: 10.0*10.0 cm Wall thickness :1.2mm Duration: 32.0s				
	Expressive Texture aims to create textured animation effects without emphasizing semantic information conveyance.		Diagonal wiping Size: 10.0*10.0 cm Wall thickness :1.2mm Duration: 29.0s				
			Radial spreading Size: 10.0*10.0 cm Wall thickness :1.2mm Duration: 28.0s				
			Growing Curve Size: 10.0*10.0 cm Wall thickness :1.2mm Duration: 25.1s				
Multi-stroke path refers to pipe structures that allow water to flow from the inlet to the outlet along multiple branching paths.	Expressive Texture The same as above.		Regular mesh Size: 10.0*10.0 cm Wall thickness :1.2mm Duration: 43.0s				
			Irregular mesh Size: 10.0*10.0 cm Wall thickness :1.2mm Duration: 75.0s				

Figure 5: Animation effects on flat surfaces. The critical color-changing temperature of the thermochromic paint applied to the surface is 31°C, while the temperature of the water flowing through is 40°C. The wall thickness and size refer to the dimensions used during 3D printing.

The detailed fabrication process involves three steps—preparation, assembly, and polishing—as illustrated in Figure 8.

5 Application Example

ViviClay displays dynamic textures driven by water flow, making it particularly suitable for water-related products such as drinking

cups, tea pets that require rinsing with tea water, and interactive water-flow decorative ornaments. Figure 1 illustrates a water cup fabricated using an assembly process, where the straight wall is printed as a flat surface with internal pipes, while the heart-shaped cup wall is printed without internal pipes. Upon assembly, the complete cup is formed. We fabricated a set of paired cups that can be

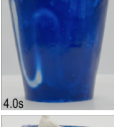
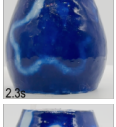
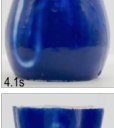
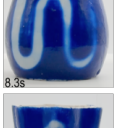
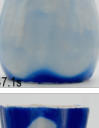
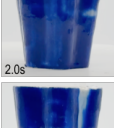
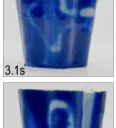
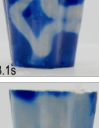
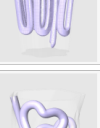
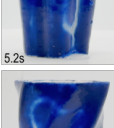
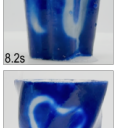
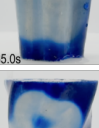
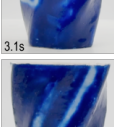
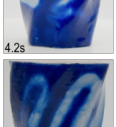
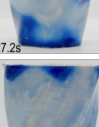
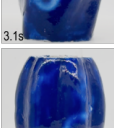
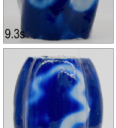
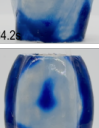
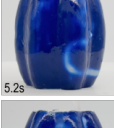
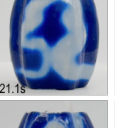
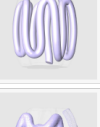
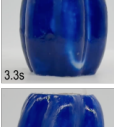
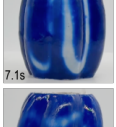
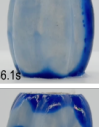
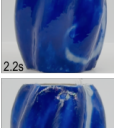
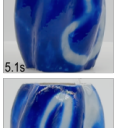
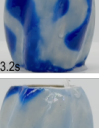
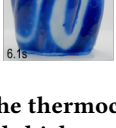
Contour	Generatrix	Twist	3D model & Specification		Animation			
Circular The horizontal contour line is circular. 	Straight line The below shape is formed by a circular contour line and a straight generatrix line. 	Cannot twist For a circular contour line, rotating it around its center does not alter the resulting shape. 		Single-stroke path icon Shamrock Size: 6.0×6.0×7.2cm Wall thickness : 1.5mm Duration: 28.5s				
				Single-stroke path texture Longitudinal wiping Size: 6.0×6.0×7.2cm Wall thickness : 1.5mm Duration: 34.1s				
	Curve The below shape is formed by a circular contour line and a curve generatrix line. 	Cannot twist The same as above. 		Single-stroke path icon Shamrock Size: 6.0×6.0×7.2cm Wall thickness : 1.5mm Duration: 11.1s				
				Single-stroke path texture Longitudinal wiping Size: 6.0×6.0×7.2cm Wall thickness : 1.5mm Duration: 47.1s				
	Straight line The below shape is formed by a non-circular contour line and a straight generatrix line. 	Not twisted The non-circular contour line does not rotate around the center point. 		Single-stroke path icon Direction Size: 6.0×6.0×7.2cm Wall thickness : 1.5mm Duration: 8.1s				
				Single-stroke path texture Longitudinal wiping Size: 6.0×6.0×7.2cm Wall thickness : 1.5mm Duration: 35.0s				
		Twisted The non-circular contour line rotate around the center point, driving changes throughout the entire shape 		Single-stroke path icon Heart Size: 6.0×6.0×7.2cm Wall thickness : 1.5mm Duration: 27.2s				
				Single-stroke path texture Longitudinal wiping Size: 6.0×6.0×7.2cm Wall thickness : 1.5mm Duration: 34.2s				
Non-circular The horizontal contour line is not circular. 	Curve The below shape is formed by a non-circular contour line and a curve generatrix line. 	Not twisted The non-circular contour line does not rotate around the center point. 		Single-stroke path icon Tree Size: 6.0×6.0×7.2cm Wall thickness : 1.5mm Duration: 31.3s				
				Single-stroke path texture Longitudinal wiping Size: 6.0×6.0×7.2cm Wall thickness : 1.5mm Duration: 36.1s				
		Twisted The non-circular contour line rotate around the center point, driving changes throughout the entire shape. 		Single-stroke path icon Cat Size: 6.0×6.0×7.2cm Wall thickness : 1.5mm Duration: 13.2s				
				Single-stroke path texture Longitudinal wiping Size: 6.0×6.0×7.2cm Wall thickness : 1.5mm Duration: 34.2s				

Figure 6: Animation effects on curved surfaces. The critical color-changing temperature of the thermochromic paint applied to the surface is 31°C, while the temperature of the water flowing through is 40°C. The wall thickness and size refer to the dimensions used during 3D printing.

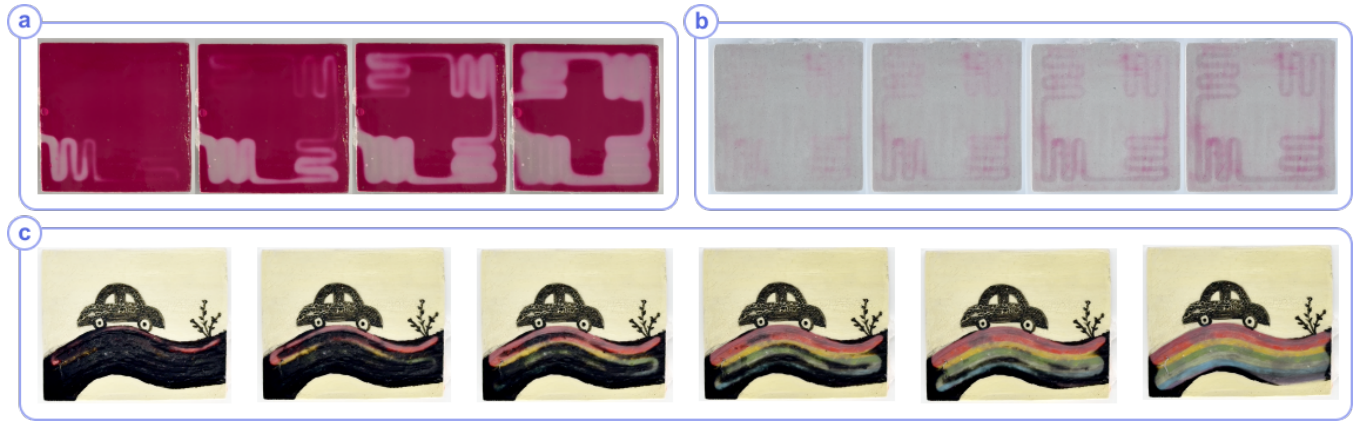


Figure 7: Two different application methods: (a,b) flat painting and (c) patterned drawing. (a) Applying thermochromic paint that transitions from colored to colorless on the ceramic surface. (b) Applying thermochromic paint that transitions from colorless to colored on the ceramic surface. (c) Drawing patterns on the ceramic surface using thermochromic paint and glaze.

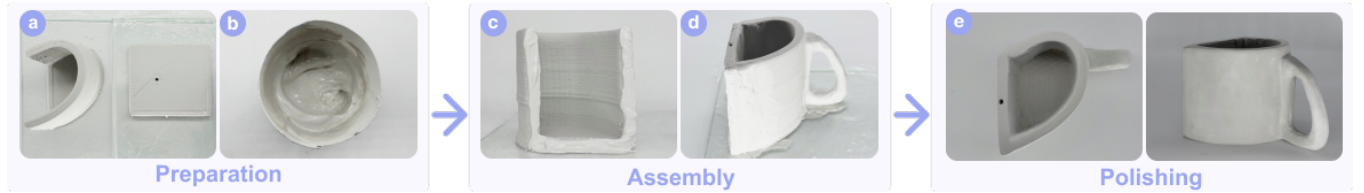


Figure 8: Detailed three steps of assembly process. (1) Preparation – includes (a) preparing the objects for assembly and (b) mixing the clay slurry to the appropriate consistency. The clay must be mixed to achieve an appropriate slurry consistency. If the slurry is too thin, the bonding will be weak; if it is too thick, it becomes difficult to apply, potentially causing gaps at the connections and increasing the risk of cracking; (2) Assembly – includes (c) evenly applying the prepared slurry to the contact surfaces and (d) assembling the parts into a compound shape; (3) Polishing – involves (e) using a damp sponge to smooth the seams, ensuring they are even and visually appealing.

combined to create a complete heart shape. A pink thermochromic paint, which transitions from colored to colorless, is uniformly applied to the surface of the straight wall. When hot water flows through the pipes, the wall dynamically displays a heart pattern, imparting a rich emotional significance to the cup. The water then flows into the cup, making it drinkable and achieving a fusion of functionality and interactive experience.

6 Conclusion and Future Work

This paper presents ViviClay, an innovative ceramic material that creates animation effects on physical surfaces by utilizing thermochromic pigments responsive to temperature changes. By integrating internal pipe structures within ceramic objects and coating their surfaces with thermochromic pigments, dynamic textures emerge as heat fluid flows through the pipes, inducing reversible color shifts. To address the challenges of clay 3D printing, such as clay’s semi-fluid nature leading to deformation and pipe blockages, experiments were conducted to refine printing techniques for flat and curved surfaces. The study proposes design strategies, printing parameter recommendations, and a design space for dynamic textures, offering technical support for diverse applications.

Future work will focus on conducting more detailed quantitative experiments on structural and printing parameters to further expand the design space, improving ViviClay’s visual performance by

addressing short animation issues caused by ceramic thermal diffusion, enhancing durability through testing and developing a design tool capable of simulating animation effects to assist creators in efficiently designing and manufacturing ViviClay. Additionally, we plan to produce more ceramic works to further explore ViviClay’s application potential in artistic ornaments and functional products.

Acknowledgments

Guanhong Liu is the corresponding author. This work was supported by Youth Fund for Humanities and Social Sciences Research, Ministry of Education 24YJCZH168. We also thank for Tianyu Yu and Danyang Song for their eagerness to help.

References

- [1] Bahare Bakhtiari, Charles Perin, and Sowmya Somanath. 2024. VISMOC: A Programmable Smocking Technique for Creating Interactive Data Physicalization. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference*. Association for Computing Machinery, New York, NY, USA, 2341–2356.
- [2] Fiona Bell, Erin McClure, Camila Friedman-Gerlicz, Ruby Ta, and Leah Buechley. 2024. Shape-Changing Clay-Dough: Taking a Material-Oriented Approach to 3D Printing Ceramic Forms. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–19.
- [3] Leah Buechley, Jaime Gould, and Fiona Bell. 2024. CeraMetal: A New Approach to Low-Cost Metal 3D Printing with Bronze Clay. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–16.

- [4] Kyung Yun Choi and Hiroshi Ishii. 2021. Therms-up!: Diy inflatables and interactive materials by upcycling wasted thermoplastic bags. In *Proceedings of the Fifteenth International Conference on Tangible, Embedded, and Embodied Interaction*. Association for Computing Machinery, New York, NY, USA, 1–8.
- [5] Camila Friedman-Gerlicz, Deanna Gelosi, Fiona Bell, and Leah Buechley. 2024. WeaveSlicer: Expanding the Range of Printable Geometries in Clay. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–16.
- [6] Devon Frost, Raina Lee, Eun-Ha Paek, and Jennifer Jacobs. 2024. SketchPath: Using Digital Drawing to Integrate the Gestural Qualities of Craft in CAM-Based Clay 3D Printing. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–16.
- [7] Ollie Hanton, Michael Wessely, Stefanie Mueller, Mike Fraser, and Anne Roudaut. 2020. ProtoSpray: Combining 3D printing and spraying to create interactive displays with arbitrary shapes. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–13.
- [8] Jean Hergel, Kevin Hinz, Sylvain Lefebvre, and Bernhard Thomaszewski. 2019. Extrusion-based ceramics printing with strictly-continuous deposition. *ACM Transactions on Graphics (TOG)* 38, 6 (2019), 1–11.
- [9] Julian Jauk, Lukas Gosch, Hana Vašatko, Markus Königsberger, Johannes Schlusche, and Milena Stavríc. 2023. Filament-reinforced 3D printing of clay. *Materials* 16, 18 (2023), 6253.
- [10] Yuhua Jin, Isabel Qamar, Michael Wessely, and Stefanie Mueller. 2020. Photochromeleon: Re-programmable multi-color textures using photochromic dyes. In *ACM SIGGRAPH 2020 Emerging Technologies*. Association for Computing Machinery, New York, NY, USA, 1–2.
- [11] Daphna Kaplan, Ronia Lencer, and Yoav Stermán. 2024. CeramiStick: a joystick controller for real-time interaction and reproduction with a ceramic 3D printer. In *Adjunct Proceedings of the 9th ACM Symposium on Computational Fabrication*. Association for Computing Machinery, New York, NY, USA, 1–3.
- [12] Guanhong Liu, Tianyu Yu, Zhihao Yao, Haiqing Xu, Yunyi Zhang, Xuhai Xu, Xiaomeng Xu, Mingyue Gao, Qirui Sun, Tingliang Zhang, et al. 2022. ViviPaint: Creating Dynamic Painting with a Thermochromic Toolkit. *Multimodal Technologies and Interaction* 6, 8 (2022), 63.
- [13] Kongpyung Moon, Zofia Marciniak, Ryo Suzuki, and Andrea Bianchi. 2024. 3D Printing Locally Activated Visual-Displays Embedded in 3D Objects via Electrically Conductive and Thermochromic Materials. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–15.
- [14] Hila Mor, Tianyu Yu, Ken Nakagaki, Benjamin Harvey Miller, Yichen Jia, and Hiroshi Ishii. 2020. Venous Materials: Towards Interactive Fluidic Mechanisms. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–14.
- [15] Takafumi Morita, Rei Sakura, Kanon Aoyama, Tomomi Imamura, Changyao Han, and Yasuaki Kakehi. 2024. Design and Fabrication for Dynamic Color-Changing on Curved 3D-Printed Surfaces. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–7.
- [16] Ilan E Moyer, Samuelle Bourgault, Devon Frost, and Jennifer Jacobs. 2024. Throwing Out Conventions: Reimagining Craft-Centered CNC Tool Design through the Digital Pottery Wheel. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–22.
- [17] Simon Olberding, Michael Wessely, and Jürgen Steimle. 2014. PrintScreen: fabricating highly customizable thin-film touch-displays. In *Proceedings of the 27th annual ACM symposium on User interface software and technology*. Association for Computing Machinery, New York, NY, USA, 281–290.
- [18] Joyce E Passananti, Ana CardenasGasca, Jennifer Jacobs, and Tobias Höllerer. 2023. SculptAR: Direct Manipulations of Machine Toolpaths in Augmented Reality for 3D Clay Printing. In *Adjunct Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. Association for Computing Machinery, New York, NY, USA, 1–3.
- [19] Parinya Punpongsanon, Xin Wen, David S Kim, and Stefanie Mueller. 2018. ColorMod: recoloring 3D printed objects using photochromic inks. In *Proceedings of the 2018 CHI Conference on human factors in computing systems*. Association for Computing Machinery, New York, NY, USA, 1–12.
- [20] Ticha Sethapakdi, Laura Huang, Vivian Hsinyueh Chan, Lung-Pan Cheng, Fernando Fuzinato Dall'Agnol, Mackenzie Leake, and Stefanie Mueller. 2023. Polygons: Designing and Fabricating Polarized Light Mosaics with User-Defined Color-Changing Behaviors. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–14.
- [21] Manlin Song, Chenyu Jia, and Katia Vega. 2018. Eunoia: Dynamically control thermochromic displays for animating patterns on fabrics. In *Proceedings of the 2018 ACM International Joint Conference and 2018 International Symposium on Pervasive and Ubiquitous Computing and Wearable Computers*. Association for Computing Machinery, New York, NY, USA, 255–258.
- [22] Manlin Song and Katia Vega. 2018. HeartMe: Thermochromic display as an expression of heart health. In *Proceedings of the 2018 ACM Conference Companion Publication on Designing Interactive Systems*. Association for Computing Machinery, New York, NY, USA, 311–314.
- [23] Cesar Torres, Jessica Chang, Advaita Patel, and Eric Paulos. 2019. Phosphenes: Crafting resistive heaters within thermoreactive composites. In *Proceedings of the 2019 on Designing Interactive Systems Conference*. Association for Computing Machinery, New York, NY, USA, 907–919.
- [24] Muhammad Umair, Corina Sas, and Miquel Alfaras. 2020. ThermoPixels: Toolkit for personalizing arousal-based interfaces through hybrid crafting. In *Proceedings of the 2020 ACM designing interactive systems conference*. Association for Computing Machinery, New York, NY, USA, 1017–1032.
- [25] Tianyang Xue, Lin Lu, Andrei Sharf, Lihao Tian, and Haochen Li. 2022. Dancing-pottery: Posture-driven pottery generative design and fabrication. In *Proceedings of the 7th Annual ACM Symposium on Computational Fabrication*. Association for Computing Machinery, New York, NY, USA, 1–11.
- [26] Tianyu Yu, Weiye Xu, Haiqing Xu, Guanhong Liu, Chang Liu, Guanyun Wang, and Haipeng Mi. 2023. Thermotion: Design and fabrication of thermofluidic composites for animation effects on object surfaces. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–19.